

Towards Evidence-based Egocentric Assistant for Clinical Perception and Action

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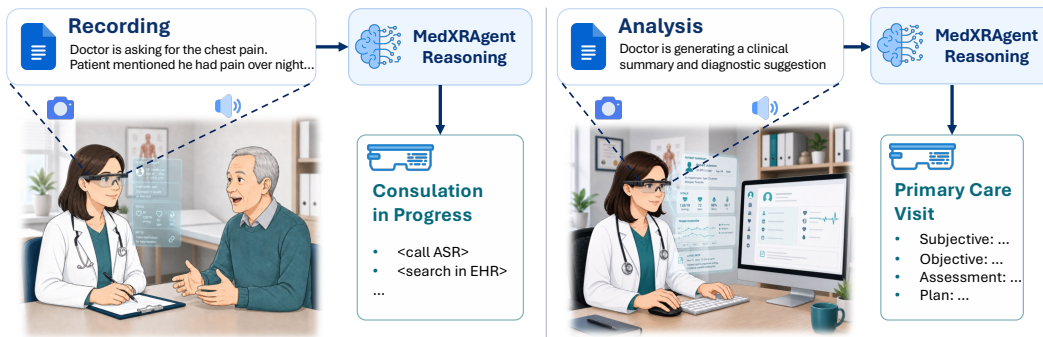


Figure 1: MedXRAgent is a context-aware MLLM agent for real-world clinical practice.

Abstract

Clinical reasoning in outpatient encounters requires dynamic integration of streamed dialogue, egocentric observations, structured patient context, EHR-derived history when available, and external medical evidence. However, most medical AI systems still perform retrospective reasoning over static prompts rather than assembling encounter context from streamed multimodal evidence. We propose MedXRAgent, an Extended Reality (XR)-supported, context-aware clinical agent that processes first-person audio and visual evidence through a perception-understanding-context-reasoning loop with web-grounded evidence retrieval. To evaluate this setting, we curate MedXR-Bench from *NEJM* Image Challenge cases, with 795 training and 271 test egocentric video cases in the benchmark manifests. MedXRAgent achieves the best open-ended diagnosis score on the test split, while post-trained Qwen3.5 variants provide strong summarization and reference-metric performance. Component ablations show that both web evidence and visual evidence provide substantial gains for open-ended diagnosis, with web evidence showing a slightly larger aggregate effect. These results support MedXRAgent as a research framework for evidence-grounded, egocentric clinical assistance rather than an autonomous diagnostic system.

1 Introduction

Clinical reasoning is a highly dynamic and longitudinal process that requires the continuous integration of evolving patient dialogue, physical observations, and historical records. During point-of-care interventions, junior physicians often face an immense cognitive load, as they must simultaneously manage patient communication, conduct physical examinations, and perform complex diagnostic

deduction [3]. While recent advances in large language models (LLMs) have demonstrated expert-level proficiency on static medical licensing exams and isolated diagnostic benchmarks, real-world clinical application remains bottlenecked by a fundamental gap between offline reasoning and situated, interactive practice. In reality, clinical reasoning is not a one-shot prediction over a fixed, fully-observed prompt; rather, it is an iterative process of hypothesis generation and context integration, where clinicians must constantly weigh the reliability of incoming multimodal signals and revise assumptions as new evidence arrives.

This continuous, evidence-gathering paradigm aligns with the principles of Evidence-Based Medicine (EBM), which dictates that clinical decisions must systematically integrate patient-specific context with critically appraised external evidence [17, 2]. However, prevailing medical AI systems fail to satisfy these requirements as they are predominantly optimized for retrospective reasoning on static datasets. By typically operating on "prepackaged" inputs, such as isolated images or curated post-hoc summaries, these systems act merely as passive post-processors. Consequently, they lack the ability to organize multi-turn encounter context, align multimodal evidence, and generate clinician-facing support within the dynamic environment of a physical consultation.

MedXRAgent addresses this gap as a context-aware clinical assistance framework for continuous reasoning over streamed encounter evidence. By integrating tool-augmented reasoning with smart-glass Extended Reality (XR) interfaces, MedXRAgent organizes doctor-patient dialogue, egocentric visual observations, EHR-derived patient context, and retrieved external evidence into an encounter context for clinician-facing decision support. Rather than functioning as a standalone diagnostic model, MedXRAgent is designed to surface structured summaries, diagnostic hypotheses, and supporting evidence while preserving clinician control during the encounter.

To rigorously evaluate MedXRAgent’s capabilities, we curate MedXR-Bench, an interactive evaluation suite derived from the NEJM Image Challenge. Unlike static VQA tasks, MedXR-Bench converts NEJM challenge’s complex clinical cases into multi-turn consultation episodes, incorporating streamed audio, egocentric visual observations, and structured EHR context. To cater to diverse deployment needs, we instantiate MedXRAgent in two variants: a cloud-based version powered by Gemini 3.1 Pro, and an open-source version designed for local deployment. For the latter, we fine-tune Qwen3.5 using task-specific Group Relative Policy Optimization (GRPO). This post-training stage improves the open-source reasoner’s ability to generate structured, evidence-grounded clinical outputs within the same MedXRAgent pipeline. Together, these components support reproducible research on clinician-facing, evidence-grounded clinical assistance rather than autonomous diagnosis.

Our primary contributions are summarized as follows:

- (1) We propose a context-aware architecture that integrates egocentric multimodal perception, structured patient context, and retrieved evidence, enabling a "perceive-reason-act" loop in real-time clinical settings via XR interfaces.
- (2) We introduce a first-of-its-kind interactive clinical benchmark that simulates the complexity of real-world consultations, moving beyond static VQA to evaluate multi-turn multimodal reasoning and diagnosis context-aware guidance.
- (3) We demonstrate the effectiveness of using GRPO to align multimodal agents with clinical reasoning standards, significantly improving the consistency and safety of diagnostic suggestions.

2 Related Work

MLLM Agents. Large multimodal models such as Flamingo [1], BLIP-2 [31], LLaVA [35], and Gemini [64] extend language-model reasoning to visual, textual, audio, and video inputs. Recent multimodal agents further use LLMs or MLLMs as controllers for tool selection, external model invocation, executable reasoning, and multi-step visual task solving, as shown by Visual ChatGPT [71], MM-ReAct [81], HuggingGPT [55], ViperGPT [60], AssistGPT [16], and LLaVA-Plus [38]. Proactive and context-aware agents additionally study assistance timing, clarification, and contextual reasoning in open-world settings, including Proactive Agent [40], proactive conversational agents [32], Ask-before-Plan [87], ContextAgent [79], and ProAgent [78]. However, these systems mainly target open-domain visual tasks, tool-use demonstrations, or daily-life assistance rather than clinical workflows, where streaming dialogue, first-person observations, patient context, retrieved evidence, and longitudinal reasoning must be integrated continuously.

XR Applications in Clinical Workflow. XR has been explored in healthcare for medical training, real-time clinical support [9], AI-XR procedural perception and action [72], and surgical applications such as planning, navigation, patient education, and rehabilitation [25]. Deployment studies further highlight barriers including usability, latency, privacy, EHR interoperability, regulation, maintenance, and reimbursement [42]. Despite these advances, existing clinical XR systems remain centered on training, visualization, navigation, or procedure-specific support, whereas MedXRAgent targets continuous outpatient reasoning from first-person observations, doctor–patient dialogue, patient history, and retrieved evidence.

Medical MLLM Benchmarks and Agentic Systems. Medical multimodal AI has been studied through medical VQA [84], VQA-RAD [27], PathVQA [18], SLAKE [34], PMC-VQA [86], and broader benchmarks for generalist biomedical AI, medical multimodal understanding, spectrum-level MLLM evaluation, and expert-level clinical reasoning [67, 10, 36, 89]. Recent medical agentic systems explore multi-agent collaboration [63, 24], multimodal tool use [29], radiology-specific reasoning [15], multimodal medical agentic intelligence [11], simulated clinical interaction [50], and guideline-based evidence-oriented diagnosis [70, 58, 17]. Despite these advances, existing benchmarks and agentic systems still mainly focus on static images, prepackaged cases, domain-specific visual reasoning, or episodic simulations, leaving continuous egocentric clinical reasoning over streaming dialogue, first-person observations, patient context, and retrieved evidence largely unexplored.

3 Context-Aware Agent Task in Clinical Perception and Action

3.1 Task Definition

Building on context-aware agents and interactive clinical agent benchmarks [79, 50], and following the evidence-based medicine principle of integrating patient-specific context with external evidence [17, 2], we define MedXRAgent as an online context-aware clinical reasoning task rather than a static medical question-answering problem. XR/AR devices function as the modality-rich sensing and interaction interfaces for the system. At each time step t , the agent receives incremental encounter observations and composes the available clinical context as:

$$x_t = (a_t, v_t), \quad C_t = \mathcal{C}(a_{\leq t}, v_{\leq t}^{\text{sel}}, P_t, w_{\leq t}), \quad Y_t = F_\theta(C_t). \quad (1)$$

Here, a_t denotes the current segment of doctor–patient audio, v_t denotes egocentric visual frames, $v_{\leq t}^{\text{sel}}$ denotes the selected clinically relevant visual evidence up to time t , P_t denotes the EHR-derived patient context available at time t , and $w_{\leq t}$ denotes external evidence retrieved up to time t . In MedXR-Bench, P_t is fixed for each case because the patient history is provided as structured case context; in deployment, it can be refreshed through EHR queries as the encounter unfolds. C_t is therefore an organized encounter context assembled from observed and retrieved evidence, rather than a learned latent memory state or a raw accumulated prompt.

This formulation reflects outpatient encounters where information is incomplete, arrives asynchronously, and requires clinical evidence to be integrated over time. The task necessitates temporal continuity across fragmented utterances and first-person observations, rather than treating the case as a single-turn multimodal prompt.

MedXR-Bench evaluates the final accumulated outputs $Y_T = (S_T, Z_T)$ at the terminal step T . **Task 1: Factual Clinical Summarization** (S_T) requires generating a concise summary of symptoms, timelines, medications, and laboratory findings. **Task 2: Open-ended Diagnostic Reasoning** (Z_T) evaluates the generation of etiologic or diagnostic conclusions grounded in the multimodal evidence. While these outputs can be rendered as real-time AR guidance in the live system, the benchmark focuses on the accuracy and clinical plausibility of the generated summary and diagnostic answers.

3.2 Task Construction

We instantiate MedXR-Bench by expanding *NEJM* Image Challenge cases into controlled outpatient-style episodes with egocentric multimodal traces [66]. Each source case provides clinically authored patient information, disease-relevant visual evidence, a diagnostic or etiologic question, answer options, a correct answer, and an explanatory rationale. Rather than preserving the original static

multiple-choice format, which can hide reasoning failures in NEJM-style medical image challenges [20], we use these fields to construct an online multimodal clinical reasoning benchmark.

For each case, we parse the patient description, diagnostic question, visual evidence, answer-related metadata, and disease-category label. We then construct a temporally ordered trace containing doctor-patient dialogue, synthesized clinical audio (via Gemini 3 Pro Audio), egocentric-style visual observations (via capturing role-play video by smart glasses), structured patient context, and targets for clinical summarization and open-ended diagnosis or etiology prediction. This design follows recent clinical-agent and diagnostic-dialogue evaluations that move beyond static QA toward patient interaction, incomplete information, and sequential reasoning [50, 68]. The trace is presented as incremental multimodal evidence rather than a fully packaged prompt, so the benchmark tests whether an agent can reason over clinical evidence as it unfolds. The AR-based egocentric interface preserves this continuous trace without disrupting routine clinical workflow. As the clinician moves across consultation, bedside examination, or patient repositioning, the sensor remains aligned with the physician’s first-person view, making the doctor’s observed evidence directly available to the agent. In this sense, egocentric perception provides a clinician-aligned evidence channel: what the physician sees during the encounter is captured as the agent’s visual context.

To avoid multiple-choice shortcuts and source-answer leakage, the original answer options, correct answer, and source explanation are excluded from all model-facing inputs and retrieval queries. These fields are retained only as offline metadata: the patient description supports the summarization target, while the correct answer defines the open-ended diagnostic or etiologic target. The processed corpus contains 1,066 cases, with 795 training and 271 test cases; Figure 2 summarizes disease categories, image modalities, dialogue turns, and audio duration.

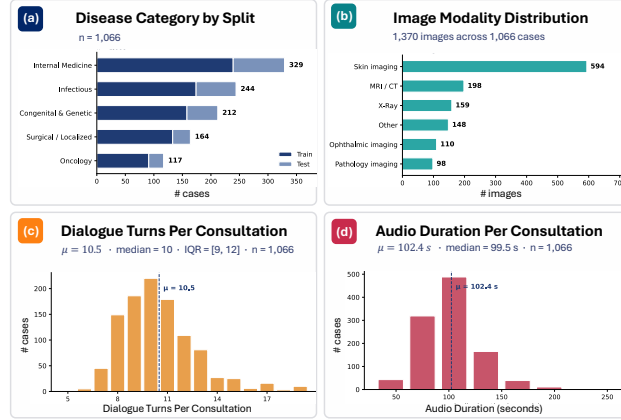


Figure 2: MedXR-Bench dataset statistics. The benchmark spans diverse disease categories, image modalities, multi-turn clinical dialogues, and audio-duration distributions.

4 MedXRAgent

4.1 System Overview

MedXRAgent is a context-aware clinical agent for continuous reasoning over streaming multimodal encounter evidence. As illustrated in Figure 3, the system processes egocentric audio and video from AR glasses together with structured patient context, constructs an evolving encounter representation, performs RAG-guided web retrieval, and generates clinician-facing support in real time. Unlike static medical QA pipelines, MedXRAgent forms an online context-aware perception-understanding-context-reasoning loop that updates its encounter state as new evidence arrives and actively orchestrates multimodal context, patient history, and external retrieval for clinical support. In live AR use, the generated summaries, suspected diagnoses, and evidence traces can be rendered to the clinician, allowing the clinician to adapt subsequent questioning or examination while remaining in control of the encounter.

The retrieval design follows evidence-based medicine [17, 49], which emphasizes integrating patient-specific context, clinical expertise, and critically appraised external evidence [2, 17, 49]. Closed medical corpora, including textbooks, disease knowledge bases, and guideline-like resources [58], provide reliable clinical priors, but medical evidence changes over time [54] and may require living updates to preserve currency and clinical utility [14]. MedXRAgent therefore couples RAG-style disease-candidate retrieval [28] with web search: RAG anchors reasoning in established disease knowledge through explicit retrieval [22, 21], while web retrieval expands evidence coverage and

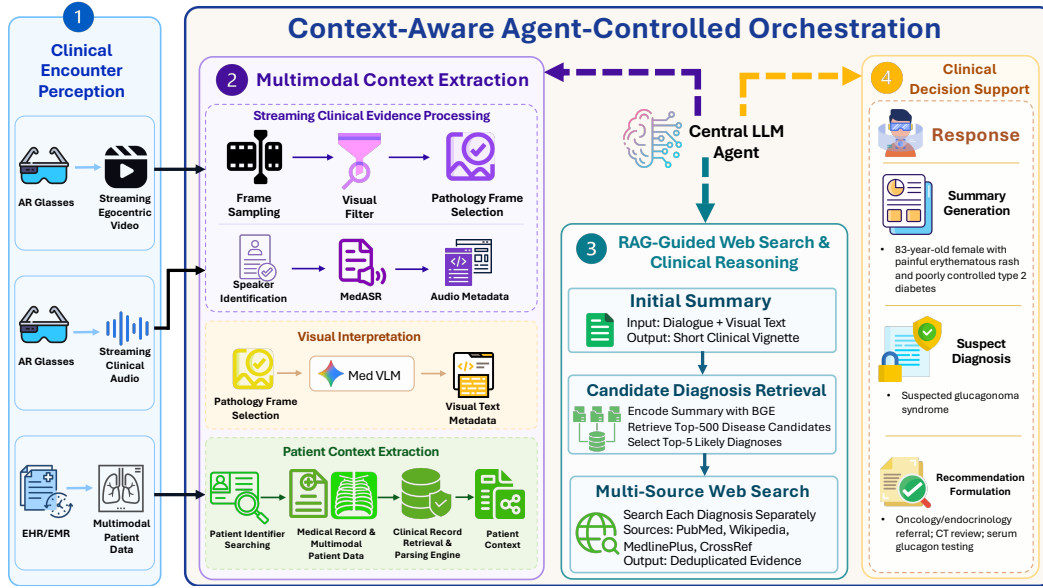


Figure 3: Overview of MedXRAgent. The system converts multimodal encounter inputs from AR glasses and structured patient context into organized clinical evidence, performs context-aware clinical reasoning with web retrieval, and produces clinician-facing outputs for AR-assisted decision support.

timeliness with source-aware external information. This design also aligns with recent medical agent and medical RAG systems that use retrieval to support evidence-oriented clinical reasoning [70, 75].

Operationally, MedXRAgent converts raw audio, video, and patient-context inputs into structured clinical evidence, including reconstructed dialogue, selected visual findings, visual-text metadata, and structured patient history. These streams are organized into a longitudinal patient state, which a central clinical agent uses for summarization, diagnosis, RAG-guided web retrieval, and compact AR-facing outputs. These outputs are framed as clinician-facing decision support rather than autonomous diagnostic decisions. This modular design follows context-aware agent pipelines [79] that separate context construction from downstream reasoning, while specializing the full loop for longitudinal clinical decision support.

4.2 Streaming Clinical Evidence Processing

Audio processing. The audio stream is processed into clinically usable dialogue rather than plain transcription. MedXRAgent combines a full-audio ASR pass with shorter speaker-labeled segments for doctor-patient attribution, using medical conversation transcription [57] and speaker-verification techniques [13] to preserve both clinical content and speaker roles. The central agent then aggregates these intermediate outputs into a temporally coherent structured dialogue, which serves as the audio-derived representation for downstream inference rather than an oracle transcript.

Visual evidence selection. The video stream is processed as a sparse but targeted evidence channel. MedXRAgent samples egocentric frames, filters for clinically relevant pathology captures, and ranks retained frames by classifier confidence, temporal diversity, and near-duplicate suppression. This design keeps a compact set of informative visual observations instead of passing the full video stream to the central model, reflecting the redundancy and computational cost of dense visual inputs in VLM inference [80]. Selected frames are then used directly as visual evidence or converted into visual-text metadata by a medical VLM [30, 52, 44].

4.3 Agent Orchestration and Web-grounded Clinical Reasoning

The central clinical agent serves as the orchestration layer over the organized encounter context, routing reconstructed dialogue, selected visual evidence or visual-text metadata, structured patient context, and external retrieval into task-specific reasoning steps. It uses this context for two coupled

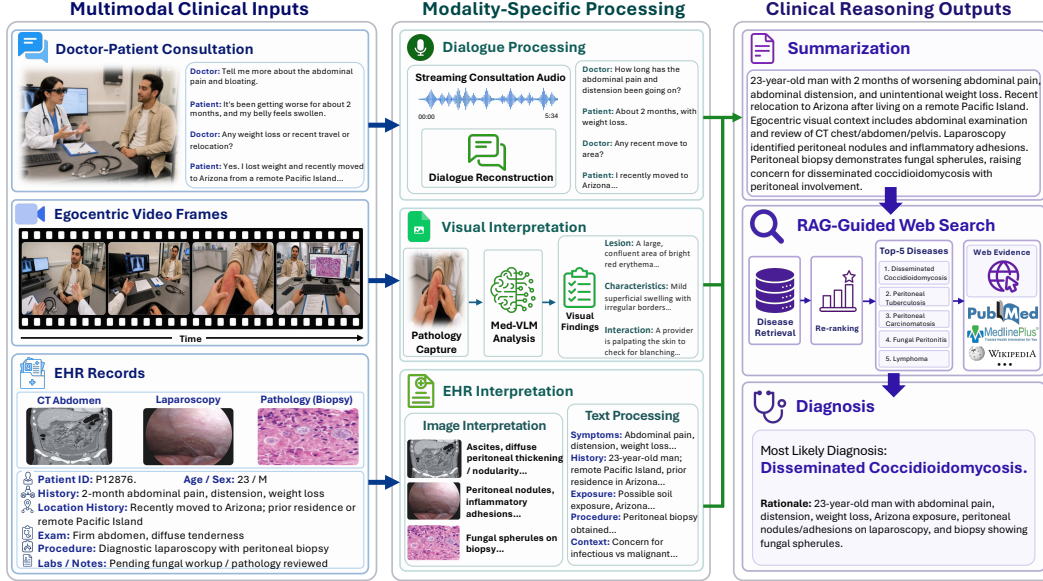


Figure 4: Example MedXRAgent workflow for an egocentric clinical encounter. The example illustrates how multimodal clinical inputs are processed into modality-specific evidence and integrated into summarization, retrieval, and diagnosis outputs.

objectives: concise visit summarization and open-ended diagnostic reasoning. To ground diagnosis without relying on free-form symptom queries, MedXRAgent uses a disease-candidate-first retrieval strategy. It first generates a no-web clinical vignette from the current encounter context, embeds it with BGE [73], and retrieves a high-recall pool of 500 candidate diseases from the medical knowledge base using FAISS [21] and dense retrieval [22]. A lightweight GPT-based reranker then selects the five most plausible disease names as focused web-search queries. This is a precision-oriented query-selection step, not a factual validator or final diagnostic model: the retriever maximizes candidate recall, while the reranker narrows the disease list for downstream search, following the retrieve-then-rerank paradigm [43] and recent LLM-based reranking methods [59, 41, 83, 47, 46].

The web backend searches each selected disease separately, deduplicates retrieved results, and serializes compact source-aware evidence blocks for the central agent. This filtering step reduces noisy or irrelevant retrieved context, which can degrade long-context and RAG performance [37, 4], and is consistent with corrective and self-reflective RAG methods that assess retrieved evidence before generation [77, 6]. Final summary and diagnosis are produced only after the central agent jointly reasons over the encounter evidence, structured patient context, and retrieved source-aware evidence. This design aligns with recent medical RAG studies and evidence-based medical-agent systems that use retrieval for diagnosis-oriented information seeking [70, 75, 74].

4.4 Post-Training for the Open-Source Version

To make MedXRAgent reproducible beyond proprietary central models, we use reinforcement learning to post-train an open-source Qwen3.5-based central reasoner that can be inserted into the same perception, patient-context, retrieval, and response-generation pipeline [48]. This stage aligns the model with MedXRAgent’s evidence-grounded diagnosis objective: given reconstructed dialogue, selected visual evidence, structured patient context, and retrieved source-aware evidence, the model should produce clinically supported open-ended diagnostic answers.

We use Group Relative Policy Optimization (GRPO) [53], a critic-free reinforcement-learning method that estimates relative advantages from groups of sampled completions. For each diagnosis prompt x , the policy samples G candidate answers $\{y_1, \dots, y_G\}$, scores each answer with the clinical reward $r(\cdot)$, and computes a mean-centered group advantage $A_i = r_i - G^{-1} \sum_{j=1}^G r_j$. We omit variance normalization because many diagnosis groups have near-zero reward variance, where dividing by a small standard deviation can amplify stochastic noise. We sample $G = 16$ completions, apply LoRA

adapters (Rank = 256) for parameter-efficient post-training [19], and use asymmetric clipping with $\epsilon = 0.2$ and $\epsilon^+ = 0.28$, following clipped policy optimization in PPO [51] and DAPO [82].

The reward function uses Gemini Pro as a structured clinical judge while following the same safety-aware rubric format used in final benchmarking. We use Gemini Pro for training-time judging because the strongest proprietary MedXRAgent pipeline also uses Gemini Pro as its central reasoner, so the reward acts as a lightweight distillation signal from the strongest pipeline into the open-source reasoner. Each independent judge call compares the prediction $\hat{y}$, reference diagnosis y^* , and evidence context E , and returns diagnosis correctness, clinical relevance, and a critical-failure flag:

$$(\text{dc}^{(k)}, \text{cr}^{(k)}, \text{cf}^{(k)}) = \text{Judge}_{\text{Gemini-3-Pro}}^{(k)}(\hat{y}, y, E), \quad k = 1, \dots, M.$$

Here, $\text{dc}^{(k)}$ measures clinical equivalence to the reference diagnosis, $\text{cr}^{(k)}$ measures whether the prediction is clinically relevant and supported by the evidence context, and $\text{cf}^{(k)}$ indicates a clinically critical failure. Writing $r_k = r^{(k)}(\hat{y}, y, E)$, the per-call reward and final averaged reward are

$$r_k = \begin{cases} 0, & \text{cf}^{(k)} = \text{True}, \\ \frac{\text{dc}^{(k)} + \text{cr}^{(k)}}{K S_{\max}}, & \text{otherwise}, \end{cases} \quad r(\hat{y}, y, E) = \frac{1}{M} \sum_{k=1}^M r_k.$$

In our implementation, $M = 5$, $K = 2$, and $S_{\max} = 5$. Thus, non-critical predictions receive the normalized average of diagnosis correctness and clinical relevance across five judge calls, while judge calls flagged as clinically critical failures contribute zero to the averaged reward. Using the same rubric structure as the benchmark evaluator helps align the training reward with the final evaluation protocol, while the final reported benchmark scores are still computed with GPT-5.4-mini. This design is supported by LLM-as-Judge studies on structured GPT-based evaluation [39, 88], healthcare benchmarks with physician-written rubrics and model-based grading [5], medical LLM-as-Judge validation against clinician evaluators [12], and clinical generation work showing that surface-level metrics miss correctness, completeness, fabricated information, and potential medical harm [69, 61].

We release the seed-controlled GRPO recipe, reward plugin, LoRA adapter, and merged Qwen3.5 checkpoint for reproducible deployment.

5 Experiments and Results

5.1 Experimental Setup

We evaluate MedXRAgent on the MedXR-Bench test split using the two final outputs defined in Section 3: clinical note summarization and open-ended diagnosis or etiology generation. For summarization, we report BLEU [45], ROUGE-L [33], BERTScore [85], MEDCON [69], and an LLM-as-Judge clinical summary score. BLEU and ROUGE-L capture lexical overlap, BERTScore captures semantic similarity, and MEDCON captures clinical concept overlap. Because reference-based metrics can miss factuality, omissions, clinical usefulness, and safety-sensitive errors in medical generation [69, 62], we use a rubric-based LLM-as-Judge protocol [39, 88, 23], following healthcare evaluation studies with clinician- or physician-grounded rubrics [12, 5, 7]. For open-ended diagnosis, the LLM-as-Judge Open-End QA score evaluates clinical equivalence to the reference answer, support from encounter evidence, and critical contradictions or unsafe failures. For both judge-based metrics, GPT-5.4-mini performs five independent judge calls per sample; we normalize scores to $[0, 1]$ and report the median to reduce judge stochasticity.

We compare four groups of methods: medical multimodal models, including Lingshu-32B [76], MedGemma-1.5-4B [52], LLaVA-Med-7B [30], MedVLM-R1 [44], and Meissa-4B [11]; medical agents, including MedAgent-Pro [70] and MDAgents [24]; general and proprietary multimodal systems, including GPT-5.4 with GPT-4o-audio, Gemini-3.1-Pro, Gemma4-31B-it [65], and Qwen3.5-27B [48]; and MedXRAgent variants with Gemini-3.1-Pro, Qwen3.5-9B, or Qwen3.5-2B as the central reasoner. Inference for all open-source model variants is conducted on NVIDIA H200 GPUs to ensure consistent benchmarking. For the interactive clinical simulations, the RayNeo X3 Pro serves as the primary AR hardware platform for real-time sensing, visualization, and user interaction.

Table 1: Main results on MedXR-Bench test split. Bold indicates the best result and underline indicates the second-best result. “Summ.” and “Open-End QA” are LLM-as-Judge scores for clinical summarization and open-ended diagnosis.

Group	Method	Centered LLM	Clinical Note Summary				Diagnosis	
			BLEU	ROUGE-L	BERTScore	MEDCON	Summ.	Open-End QA
Medical VLMs	MedVLM-R1	Qwen2-VL-2B	0.089	0.353	0.418	0.383	0.389	0.115
	LLaVA-Med-7B	Mistral-7B	0.003	0.077	0.011	0.014	0.362	0.165
	MedGemma-1.5-4B	Gemma 3-4B	0.048	0.205	0.056	0.306	0.497	0.183
	Meissa-4B	Qwen3-VL-4B	0.165	0.478	0.554	0.474	0.553	0.208
	Lingshu-32B	Qwen2.5VL-32B	0.250	0.570	0.640	0.567	0.628	0.375
Medical agents	MDAgents (MedASR)	GPT-4o	N/A	N/A	N/A	N/A	0.488	0.376
	MDAgents (Gemini ASR)	GPT-4o	N/A	N/A	N/A	N/A	0.480	0.398
	MedAgent-Pro (Gemini ASR)	GPT-4o	0.118	0.428	0.515	0.418	0.593	0.568
	MedAgent-Pro (MedASR)	GPT-4o	0.139	0.444	0.531	0.451	0.599	0.592
General MLLMs	Qwen3.5-27B	-	0.134	0.310	0.301	0.366	0.595	0.384
	Gemma4-31B-it	-	0.290	0.586	0.663	0.607	0.662	0.582
	Gemini-3.1-Pro (direct)	-	0.285	0.582	0.649	0.571	0.524	0.710
	GPT-5.4 + GPT-4o-audio	-	0.232	0.516	0.579	0.520	0.582	0.743
Ours	MedXRAgent	Qwen3.5-2B (GRPO)	<u>0.370</u>	<u>0.653</u>	<u>0.734</u>	<u>0.674</u>	<u>0.725</u>	0.607
		Qwen3.5-9B (GRPO)	0.415	0.687	0.761	0.691	0.794	<u>0.775</u>
		Gemini-3.1-Pro	0.240	0.559	0.613	0.552	0.641	0.808

Table 2: Evidence-channel ablation of the Gemini-centered MedXRAgent pipeline. Removing web or visual evidence substantially reduces Open-End QA; parenthesized values denote changes relative to the full pipeline.

Configuration	BLEU	ROUGE-L	BERTScore	MEDCON	Summ.	Open-End QA
Full pipeline	0.240	0.559	0.613	0.552	0.641	0.808
– web evidence	0.244 (+0.004)	0.561 (+0.002)	0.613 (+0.000)	0.553 (+0.001)	0.611 (−0.030)	0.705 (− 0.103)
– visual evidence	0.240 (+0.000)	0.561 (+0.002)	0.614 (+0.001)	0.557 (+0.005)	0.617 (−0.024)	0.712 (− 0.096)

5.2 Main Results

Table 1 reports the main results on the MedXR-Bench test split. MedXRAgent with Gemini-3.1-Pro achieves the highest Open-End QA score, reaching 0.808 and outperforming all baselines. MedXRAgent with Qwen3.5-9B after GRPO post-training achieves the best summarization performance across traditional NLP metrics, and the LLM-as-Judge summary score, while also obtaining the second-best Open-End QA score. This task split reflects the different nature of the two outputs: clinical summarization is more format- and reference-aligned, making it more amenable to task-specific post-training [69, 7], whereas open-ended diagnosis is more knowledge-intensive and evidence-grounded, requiring broader clinical knowledge and multimodal reasoning [56, 68, 70, 74].

Standalone medical VLMs perform substantially worse on open-ended diagnosis, indicating that static medical image-text reasoning does not directly transfer to longitudinal clinical reasoning over dialogue, visual observations, patient context, and retrieved evidence. Medical agent baselines improve over most standalone VLMs, but remain below MedXRAgent. This gap suggests that generic tool use alone is insufficient for MedXR-Bench; strong performance requires encounter-specific context construction, evidence selection, and diagnosis-oriented retrieval.

5.3 Ablation Study

Evidence-channel ablations. Starting from the full Gemini-centered pipeline, we ablate web evidence and visual evidence separately while keeping the remaining inputs and the central reasoner fixed. Table 2 reports the resulting changes. Removing web evidence causes the largest degradation in Open-End QA, reducing the score from 0.808 to 0.705, a 12.7% relative drop. By contrast, its effect on summarization is smaller, with the summary judge score decreasing from 0.641 to 0.611. BLEU, ROUGE-L, BERTScore, and MEDCON remain nearly unchanged, suggesting that web evidence mainly supports diagnosis-oriented clinical reasoning rather than surface-level textual overlap. This is consistent with prior medical RAG and evidence-grounded medical-agent studies [74, 70]. Removing visual evidence also lowers performance, reducing Open-End QA from 0.808 to 0.712 and the summary judge score from 0.641 to 0.617. Under fixed dialogue and patient context, web evidence primarily provides external medical knowledge that is especially useful for open-ended diagnostic

answering, whereas selected visual evidence contributes complementary morphology and image findings that are not fully recoverable from dialogue alone.

GRPO post-training ablation. To isolate the effect of GRPO [53], we compare zero-shot and GRPO-post-trained Qwen3.5-9B central reasoners under the same MedXRAgent pipeline, with the Gemini-centered system reported only as a reference. As shown in Table 3, GRPO improves every metric for Qwen3.5-9B: Open-End QA increases from 0.572 to 0.775, and the summary judge score increases from 0.537 to 0.794. It also substantially improves BLEU, ROUGE-L, BERTScore, and MEDCON, indicating stronger alignment with reference clinical notes and better preservation of clinical content. Notably, the GRPO-trained Qwen reasoner exceeds the Gemini reference on summarization-oriented metrics, but remains slightly lower on Open-End QA. This pattern is expected: clinical note summarization is more schema- and reference-aligned, allowing GRPO to adapt the model to the desired note structure and content selection, whereas open-ended diagnostic QA relies more heavily on broad medical knowledge and diagnostic reasoning capacity, where the Gemini-centered reasoner remains stronger. Overall, these results suggest that task-specific post-training can make an open-source central reasoner competitive within the same evidence-grounded pipeline, consistent with recent reinforcement-learning approaches for medical reasoning [44], while stronger base-model knowledge remains important for diagnostic answering.

Table 3: Effect of GRPO fine-tuning on the MedXRAgent Central LLM (Qwen3.5-9B). MedXRAgent (Gemini) numbers are shown for reference.

Metric	Qwen3.5-9B			MedXRAgent (Gemini)
	Zero-shot	+ GRPO	Rel. gain	
BLEU	0.111	0.415	+274%	0.240
ROUGE-L	0.423	0.687	+62%	0.559
BERTScore	0.450	0.761	+69%	0.613
MEDCON	0.418	0.691	+65%	0.552
Summarization	0.537	0.794	+48%	0.641
Open-End QA	0.572	0.775	+35%	0.808

5.4 Category-Level and Failure Analysis

Category-level results show that MedXRAgent’s gains are uneven across clinical domains. MedXRAgent with Gemini-3.1-Pro performs strongest on internal/systemic, infectious, and congenital/genetic cases, while the GRPO-trained Qwen3.5-9B variant performs best on surgical/localized cases. GPT-5.4 with GPT-4o-audio remains strongest on oncology, indicating that tumor-related cases remain challenging for the full MedXRAgent pipeline. This pattern suggests that strong aggregate diagnosis performance does not eliminate category-specific weaknesses, especially in cases requiring fine-grained pathological, oncologic, or rare-disease reasoning [8, 20].

Remaining failures mainly arise when decisive evidence is subtle, weakly captured, or under-specified in the retained encounter context. Pathology-heavy and oncology cases can require visual details beyond the selected egocentric frames; retrieval can introduce broad, overlapping, or distracting evidence [4, 37, 74]; and the final reasoner may over-weight plausible but non-specific evidence when the encounter trace is incomplete. Representative cases further show that retrieval, visual evidence, and patient history are most useful when they provide a decisive clue, but they do not guarantee robustness when the relevant clue is missing or ambiguous. These failure modes reinforce the intended role of MedXRAgent as clinician-facing decision support rather than an autonomous diagnostic system [26].

6 Discussion and Conclusion

We presented MedXRAgent, an XR-supported egocentric clinical agent that moves medical AI beyond static question answering toward continuous, evidence-grounded assistance at the point of care. By coupling streaming multimodal perception with organized encounter context and retrieval-guided reasoning, MedXRAgent and MedXR-Bench provide a first step for studying clinical agents that observe, integrate, retrieve, and reason over evolving encounter evidence. Our results show the promise of this paradigm and highlight the importance of evidence grounding and task-specific post-training for reliable clinical outputs. Looking forward, we envision egocentric medical agents becoming collaborative clinical copilots that organize encounter context, surface timely evidence, and reduce clinicians’ cognitive burden while keeping human experts in control. We hope this work helps establish a foundation for safer, more interactive, and more clinically grounded medical AI systems.

References

- [1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35: 23716–23736, 2022.
- [2] Loai Albarqouni, Tammy Hoffmann, Sharon Straus, Nina Rydland Olsen, Taryn Young, Dragan Ilic, Terrence Shaneyfelt, R Brian Haynes, Gordon Guyatt, and Paul Glasziou. Core competencies in evidence-based practice for health professionals: consensus statement based on a systematic review and delphi survey. *JAMA network open*, 1(2):e180281, 2018.
- [3] Luke N Allen, Jialing Lin, Bradley Max Segal, Kagiso Ndlovu, Davide Bilardi, and Luisa M Pettigrew. Artificial intelligence in primary care: frameworks, challenges, and guardrails. *The Lancet Primary Care*, 2(3), 2026.
- [4] Chen Amiraz, Florin Cuconasu, Simone Filice, and Zohar Karnin. The distracting effect: Understanding irrelevant passages in rag. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18228–18258, 2025.
- [5] Rahul K Arora, Jason Wei, Rebecca Soskin Hicks, Preston Bowman, Joaquin Quiñonero-Candela, Foivos Tsimpourlas, Michael Sharman, Meghan Shah, Andrea Vallone, Alex Beutel, et al. Healthbench: Evaluating large language models towards improved human health. *arXiv preprint arXiv:2505.08775*, 2025.
- [6] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*, 2023.
- [7] Suhana Bedi, Hejie Cui, Miguel Fuentes, Alyssa Unell, Michael Wornow, Juan M Banda, Nikesh Kotecha, Timothy Keyes, Yifan Mai, Mert Oez, et al. Medhelm: Holistic evaluation of large language models for medical tasks. *arXiv preprint arXiv:2505.23802*, 2025.
- [8] Manuela Benary, Xing David Wang, Max Schmidt, Dominik Soll, Georg Hilfenhaus, Mani Nassir, Christian Sigler, Maren Knödler, Ulrich Keller, Dieter Beule, et al. Leveraging large language models for decision support in personalized oncology. *JAMA Network Open*, 6(11): e2343689, 2023.
- [9] BK Burian, RE Harari, JM Robertson, D Musson, CN Pozner, T Doyle, DS Smink, P Miccile, P Paladugu, B Atamna, et al. Using extended reality (xr) for medical training and real-time clinical support during deep space missions. *Applied Ergonomics*, 106:103902, 2023.
- [10] Pengcheng Chen, Jin Ye, Guoan Wang, Yanjun Li, Zhongying Deng, Wei Li, Tianbin Li, Haodong Duan, Ziyang Huang, Yanzhou Su, et al. Gmai-mmbench: A comprehensive multimodal evaluation benchmark towards general medical ai. *Advances in Neural Information Processing Systems*, 37:94327–94427, 2024.
- [11] Yixiong Chen, Xinyi Bai, Yue Pan, Zongwei Zhou, and Alan Yuille. Meissa: Multi-modal medical agentic intelligence. *arXiv preprint arXiv:2603.09018*, 2026.
- [12] Emma Croxford, Yanjun Gao, Elliot First, Nicholas Pellegrino, Miranda Schnier, John Caskey, Madeline Oguss, Graham Wills, Guanhua Chen, Dmitriy Dligach, et al. Evaluating clinical ai summaries with large language models as judges. *npj Digital Medicine*, 8(1):640, 2025.
- [13] Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. Ecapa-tdnn: Emphasized channel attention, propagation and aggregation in tdnn based speaker verification. *arXiv preprint arXiv:2005.07143*, 2020.
- [14] Julian H Elliott, Anneliese Synnot, Tari Turner, Mark Simmonds, Elie A Akl, Steve McDonald, Georgia Salanti, Joerg Meerpohl, Harriet MacLehose, John Hilton, et al. Living systematic review: 1. introduction—the why, what, when, and how. *Journal of clinical epidemiology*, 91: 23–30, 2017.

- 409 [15] Adibvafa Fallahpour, Jun Ma, Alif Munim, Hongwei Lyu, and Bo Wang. Medrax: Medical
410 reasoning agent for chest x-ray. *arXiv preprint arXiv:2502.02673*, 2025.
- 411 [16] Difei Gao, Lei Ji, Luowei Zhou, Kevin Qinghong Lin, Joya Chen, Zihan Fan, and Mike Zheng
412 Shou. Assistgpt: A general multi-modal assistant that can plan, execute, inspect, and learn.
413 *arXiv preprint arXiv:2306.08640*, 2023.
- 414 [17] Gordon Guyatt, John Cairns, David Churchill, Deborah Cook, Brian Haynes, Jack Hirsh, Jan
415 Irvine, Mark Levine, Mitchell Levine, Jim Nishikawa, et al. Evidence-based medicine: a new
416 approach to teaching the practice of medicine. *jama*, 268(17):2420–2425, 1992.
- 417 [18] Xuehai He, Yichen Zhang, Luntian Mou, Eric Xing, and Pengtao Xie. Pathvqa: 30000+
418 questions for medical visual question answering. *arXiv preprint arXiv:2003.10286*, 2020.
- 419 [19] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang
420 Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3,
421 2022.
- 422 [20] Qiao Jin, Fangyuan Chen, Yiliang Zhou, Ziyang Xu, Justin M Cheung, Robert Chen, Ronald M
423 Summers, Justin F Rousseau, Peiyun Ni, Marc J Landsman, et al. Hidden flaws behind
424 expert-level accuracy of multimodal gpt-4 vision in medicine. *NPJ Digital Medicine*, 7(1):190,
425 2024.
- 426 [21] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *IEEE*
427 *transactions on big data*, 7(3):535–547, 2019.
- 428 [22] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov,
429 Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering.
430 In *Proceedings of the 2020 conference on empirical methods in natural language processing*
431 *(EMNLP)*, pages 6769–6781, 2020.
- 432 [23] Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo
433 Yun, Seongjin Shin, Sungdong Kim, James Thorne, et al. Prometheus: Inducing fine-grained
434 evaluation capability in language models. In *The Twelfth International Conference on Learning*
435 *Representations*, 2023.
- 436 [24] Yubin Kim, Chanwoo Park, Hyewon Jeong, Yik S Chan, Xuhai Xu, Daniel McDuff, Hyeon-
437 hoon Lee, Marzyeh Ghassemi, Cynthia Breazeal, and Hae W Park. Mdagents: An adaptive
438 collaboration of llms for medical decision-making. *Advances in Neural Information Processing*
439 *Systems*, 37:79410–79452, 2024.
- 440 [25] Mikolaj R Kowal, Thomas Williams, Alexios Dosis, Samir Pathak, Shahid Farid, Deborah D
441 Stocken, Peter Lodge, Sharib Ali, Damian Tolan, and David G Jayne. General surgery 4.0—a
442 systematic review of extended reality interventions in general surgery. *Surgical Innovation*,
443 page 15533506251415440, 2026.
- 444 [26] Steven Labkoff, Bilikis Oladimeji, Joseph Kannry, Anthony Solomonides, Russell Leftwich,
445 Eileen Koski, Amanda L Joseph, Monica Lopez-Gonzalez, Lee A Fleisher, Kimberly Nolen,
446 et al. Toward a responsible future: recommendations for ai-enabled clinical decision support.
447 *Journal of the American Medical Informatics Association*, 31(11):2730–2739, 2024.
- 448 [27] Jason J Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. A dataset of
449 clinically generated visual questions and answers about radiology images. *Scientific data*, 5(1):
450 180251, 2018.
- 451 [28] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman
452 Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented
453 generation for knowledge-intensive nlp tasks. *Advances in neural information processing*
454 *systems*, 33:9459–9474, 2020.
- 455 [29] Binxu Li, Tiankai Yan, Yuanting Pan, Jie Luo, Ruiyang Ji, Jiayuan Ding, Zhe Xu, Shilong Liu,
456 Haoyu Dong, Zihao Lin, et al. Mmedagent: Learning to use medical tools with multi-modal
457 agent. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages
458 8745–8760, 2024.

- [30] Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36: 28541–28564, 2023.
- [31] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR, 2023.
- [32] Lizi Liao, Grace Hui Yang, and Chirag Shah. Proactive conversational agents. In *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining*, pages 1244–1247, 2023.
- [33] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.
- [34] Bo Liu, Li-Ming Zhan, Li Xu, Lin Ma, Yan Yang, and Xiao-Ming Wu. Slake: A semantically-labeled knowledge-enhanced dataset for medical visual question answering. In *2021 IEEE 18th international symposium on biomedical imaging (ISBI)*, pages 1650–1654. IEEE, 2021.
- [35] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- [36] Jie Liu, Wenxuan Wang, Su Yihang, Jingyuan Huang, Yudi Zhang, Cheng-Yi Li, Wenting Chen, Xiaohan Xing, Kao-Jung Chang, Linlin Shen, et al. Asclepius: A spectrum evaluation benchmark for medical multi-modal large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24181–24201, 2025.
- [37] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024.
- [38] Shilong Liu, Hao Cheng, Haotian Liu, Hao Zhang, Feng Li, Tianhe Ren, Xueyan Zou, Jianwei Yang, Hang Su, Jun Zhu, et al. Llava-plus: Learning to use tools for creating multimodal agents. In *European conference on computer vision*, pages 126–142. Springer, 2024.
- [39] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: Nlg evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pages 2511–2522, 2023.
- [40] Yaxi Lu, Shenzhi Yang, Cheng Qian, Guirong Chen, Qinyu Luo, Yesai Wu, Huadong Wang, Xin Cong, Zhong Zhang, Yankai Lin, et al. Proactive agent: Shifting llm agents from reactive responses to active assistance. *arXiv preprint arXiv:2410.12361*, 2024.
- [41] Xueguang Ma, Xinyu Zhang, Ronak Pradeep, and Jimmy Lin. Zero-shot listwise document reranking with a large language model. *arXiv preprint arXiv:2305.02156*, 2023.
- [42] Joseph W Morgan, Rahul A Patel, and Scott Campbell. Practical considerations of clinical xr (ar/vr) deployments. *Frontiers in Virtual Reality*, 6:1517402, 2025.
- [43] Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with bert. *arXiv preprint arXiv:1901.04085*, 2019.
- [44] Jiazhen Pan, Che Liu, Junde Wu, Fenglin Liu, Jiayuan Zhu, Hongwei Bran Li, Chen Chen, Cheng Ouyang, and Daniel Rueckert. Medvlm-r1: Incentivizing medical reasoning capability of vision-language models (vlms) via reinforcement learning. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 337–347. Springer, 2025.
- [45] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.

- [46] Ronak Pradeep, Sahel Sharifymoghaddam, and Jimmy Lin. Rankvicuna: Zero-shot listwise document reranking with open-source large language models. *arXiv preprint arXiv:2309.15088*, 2023.
- [47] Zhen Qin, Rolf Jagerman, Kai Hui, Honglei Zhuang, Junru Wu, Le Yan, Jiaming Shen, Tianqi Liu, Jialu Liu, Donald Metzler, et al. Large language models are effective text rankers with pairwise ranking prompting. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 1504–1518, 2024.
- [48] Qwen Team. Qwen3.5: Towards native multimodal agents, February 2026. URL <https://qwen.ai/blog?id=qwen3.5>.
- [49] David L Sackett, William MC Rosenberg, JA Muir Gray, R Brian Haynes, and W Scott Richardson. Evidence based medicine: what it is and what it isn't, 1996.
- [50] Samuel Schmidgall, Rojin Ziaei, Carl Harris, Eduardo Reis, Jeffrey Jopling, and Michael Moor. Agentclinic: a multimodal agent benchmark to evaluate ai in simulated clinical environments. *arXiv preprint arXiv:2405.07960*, 2024.
- [51] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [52] Andrew Sellergrén, Chufan Gao, Fereshteh Mahvar, Timo Kohlberger, Fayaz Jamil, Madeleine Traverse, Alberto Tono, Bashir Sadjad, Lin Yang, Charles Lau, et al. Medgemma 1.5 technical report. *arXiv preprint arXiv:2604.05081*, 2026.
- [53] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [54] Paul G Shekelle, Eduardo Ortiz, Shannon Rhodes, Sally C Morton, Martin P Eccles, Jeremy M Grimshaw, and Steven H Woolf. Validity of the agency for healthcare research and quality clinical practice guidelines: how quickly do guidelines become outdated? *Jama*, 286(12): 1461–1467, 2001.
- [55] Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. Hugginggpt: Solving ai tasks with chatgpt and its friends in hugging face. *Advances in Neural Information Processing Systems*, 36:38154–38180, 2023.
- [56] Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, et al. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180, 2023.
- [57] Hagen Soltau, Mingqiu Wang, Izhak Shafran, and Laurent El Shafey. Understanding medical conversations: Rich transcription, confidence scores & information extraction. *arXiv preprint arXiv:2104.02219*, 2021.
- [58] Earl Steinberg, Sheldon Greenfield, Dianne Miller Wolman, Michelle Mancher, and Robin Graham. *Clinical practice guidelines we can trust*. national academies press, 2011.
- [59] Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. Is chatgpt good at search? investigating large language models as re-ranking agents. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pages 14918–14937, 2023.
- [60] Dídac Surís, Sachit Menon, and Carl Vondrick. Vipergpt: Visual inference via python execution for reasoning. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 11888–11898, 2023.
- [61] Thomas Yu Chow Tam, Sonish Sivarajkumar, Sumit Kapoor, Alisa V Stolyar, Katelyn Polanska, Karleigh R McCarthy, Hunter Osterhoudt, Xizhi Wu, Shyam Visweswaran, Sunyang Fu, et al. A framework for human evaluation of large language models in healthcare derived from literature review. *NPJ digital medicine*, 7(1):258, 2024.

- [62] Liyan Tang, Zhaoyi Sun, Betina Idnay, Jordan G Nestor, Ali Soroush, Pierre A Elias, Ziyang Xu, Ying Ding, Greg Durrett, Justin F Rousseau, et al. Evaluating large language models on medical evidence summarization. *NPJ digital medicine*, 6(1):158, 2023.
- [63] Xiangru Tang, Anni Zou, Zhuosheng Zhang, Ziming Li, Yilun Zhao, Xingyao Zhang, Arman Cohan, and Mark Gerstein. Medagents: Large language models as collaborators for zero-shot medical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 599–621, 2024.
- [64] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [65] Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean bastien Grill, Sabela Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, Gaël Liu, Francesco Visin, Kathleen Kenealy, Lucas Beyer, Xiaohai Zhai, Anton Tsitsulin, Robert Busa-Fekete, Alex Feng, Naveen Sachdeva, Benjamin Coleman, Yi Gao, Basil Mustafa, Iain Barr, Emilio Parisotto, David Tian, Matan Eyal, Colin Cherry, Jan-Thorsten Peter, Danila Sinopalnikov, Surya Bhupatiraju, Rishabh Agarwal, Mehran Kazemi, Dan Malkin, Ravin Kumar, David Vilar, Idan Brusilovsky, Jiaming Luo, Andreas Steiner, Abe Friesen, Abhanshu Sharma, Abheesht Sharma, Adi Mayrav Gilady, Adrian Goedeckemeyer, Alaa Saade, Alex Feng, Alexander Kolesnikov, Alexei Bendebury, Alvin Abdagic, Amit Vadi, András György, André Susano Pinto, Anil Das, Ankur Bapna, Antoine Miech, Antoine Yang, Antonia Paterson, Ashish Shenoy, Ayan Chakrabarti, Bilal Piot, Bo Wu, Bobak Shahriari, Bryce Pettrini, Charlie Chen, Charline Le Lan, Christopher A. Choquette-Choo, CJ Carey, Cormac Brick, Daniel Deutsch, Danielle Eisenbud, Dee Cattle, Derek Cheng, Dimitris Paparas, Divyashree Shivakumar Sreepathihalli, Doug Reid, Dustin Tran, Dustin Zelle, Eric Noland, Erwin Huizenga, Eugene Kharitonov, Frederick Liu, Gagik Amirkhanyan, Glenn Cameron, Hadi Hashemi, Hanna Klimczak-Plucińska, Harman Singh, Harsh Mehta, Harshal Tushar Lehri, Hussein Hazimeh, Ian Ballantyne, Idan Szpektor, Ivan Nardini, Jean Pouget-Abadie, Jetha Chan, Joe Stanton, John Wieting, Jonathan Lai, Jordi Orbay, Joseph Fernandez, Josh Newlan, Ju yeong Ji, Jyotinder Singh, Kat Black, Kathy Yu, Kevin Hui, Kiran Vodrahalli, Klaus Greff, Linhai Qiu, Marcella Valentine, Marina Coelho, Marvin Ritter, Matt Hoffman, Matthew Watson, Mayank Chaturvedi, Michael Moynihan, Min Ma, Nabila Babar, Natasha Noy, Nathan Byrd, Nick Roy, Nikola Momchev, Nilay Chauhan, Naveen Sachdeva, Oskar Bunyan, Pankil Botarda, Paul Caron, Paul Kishan Rubenstein, Phil Culliton, Philipp Schmid, Pier Giuseppe Sessa, Pingmei Xu, Piotr Stanczyk, Pouya Tafti, Rakesh Shivanna, Renjie Wu, Renke Pan, Reza Rokni, Rob Willoughby, Rohith Vallu, Ryan Mullins, Sammy Jerome, Sara Smoot, Sertan Girgin, Shariq Iqbal, Shashir Reddy, Shruti Sheth, Siim Pöder, Sijal Bhatnagar, Sindhu Raghuram Panyam, Sivan Eiger, Susan Zhang, Tianqi Liu, Trevor Yacovone, Tyler Liechty, Uday Kalra, Utku Evci, Vedant Misra, Vincent Roseberry, Vlad Feinberg, Vlad Kolesnikov, Woohyun Han, Woosuk Kwon, Xi Chen, Yinlam Chow, Yuvein Zhu, Zichuan Wei, Zoltan Egyed, Victor Cotruta, Minh Giang, Phoebe Kirk, Anand Rao, Kat Black, Nabila Babar, Jessica Lo, Erica Moreira, Luiz Gustavo Martins, Omar Sanseviero, Lucas Gonzalez, Zach Gleicher, Tris Warkentin, Vahab Mirrokni, Evan Senter, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, Yossi Matias, D. Sculley, Slav Petrov, Noah Fiedel, Noam Shazeer, Oriol Vinyals, Jeff Dean, Demis Hassabis, Koray Kavukcuoglu, Clement Farabet, Elena Buchatskaya, Jean-Baptiste Alayrac, Rohan Anil, Dmitry, Lepikhin, Sebastian Borgeaud, Olivier Bachem, Armand Joulin, Alek Andreev, Cassidy Hardin, Robert Dadashi, and Léonard Hussenot. Gemma 3 technical report, 2025. URL <https://arxiv.org/abs/2503.19786>.
- [66] The New England Journal of Medicine. Image challenge. <https://www.nejm.org/image-challenge>. Accessed: 2026-04-23.
- [67] Tao Tu, Shekoofeh Azizi, Danny Driess, Mike Schaekermann, Mohamed Amin, Pi-Chuan Chang, Andrew Carroll, Charles Lau, Ryutaro Tanno, Ira Ktena, et al. Towards generalist biomedical ai. *Nejm Ai*, 1(3):AIoa2300138, 2024.
- [68] Tao Tu, Mike Schaekermann, Anil Palepu, Khaled Saab, Jan Freyberg, Ryutaro Tanno, Amy Wang, Brenna Li, Mohamed Amin, Yong Cheng, et al. Towards conversational diagnostic artificial intelligence. *Nature*, 642(8067):442–450, 2025.

- [69] Dave Van Veen, Cara Van Uden, Louis Blankemeier, Jean-Benoit Delbrouck, Asad Aali, Christian Bluethgen, Anuj Pareek, Malgorzata Polacin, Eduardo Pontes Reis, Anna Seehofnerová, et al. Adapted large language models can outperform medical experts in clinical text summarization. *Nature medicine*, 30(4):1134–1142, 2024.
- [70] Ziyue Wang, Junde Wu, Linghan Cai, Chang Han Low, Xihong Yang, Qiaxuan Li, and Yueming Jin. Medagent-pro: Towards evidence-based multi-modal medical diagnosis via reasoning agentic workflow. *arXiv preprint arXiv:2503.18968*, 2025.
- [71] Chenfei Wu, Shengming Yin, Weizhen Qi, Xiaodong Wang, Zecheng Tang, and Nan Duan. Visual chatgpt: Talking, drawing and editing with visual foundation models. *arXiv preprint arXiv:2303.04671*, 2023.
- [72] Yingcheng Charles Wu, Ming Yin, Baiyu Shi, Zaixi Zhang, Di Yin, Xiaotong Wang, Youjuan Wang, Jigang Fan, Ruofan Jin, Hanchen Wang, et al. Medos: Ai-xr-cobot world model for clinical perception and action. *medRxiv*, pages 2026–02, 2026.
- [73] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-pack: Packed resources for general chinese embeddings. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, pages 641–649, 2024.
- [74] Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6233–6251, 2024.
- [75] Guangzhi Xiong, Qiao Jin, Xiao Wang, Minjia Zhang, Zhiyong Lu, and Aidong Zhang. Improving retrieval-augmented generation in medicine with iterative follow-up questions. In *Biocomputing 2025: Proceedings of the Pacific Symposium*, pages 199–214. World Scientific, 2024.
- [76] Weiwen Xu, Hou Pong Chan, Long Li, Mahani Aljunied, Ruifeng Yuan, Jianyu Wang, Chenghao Xiao, Guizhen Chen, Chaoqun Liu, Zhaodonghui Li, et al. Lingshu: A generalist foundation model for unified multimodal medical understanding and reasoning. *arXiv preprint arXiv:2506.07044*, 2025.
- [77] Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. 2024.
- [78] Bufang Yang, Lilin Xu, Liekang Zeng, Yunqi Guo, Siyang Jiang, Wenrui Lu, Kaiwei Liu, Hancheng Xiang, Xiaofan Jiang, Guoliang Xing, et al. Proagent: Harnessing on-demand sensory contexts for proactive llm agent systems. *arXiv preprint arXiv:2512.06721*, 2025.
- [79] Bufang Yang, Lilin Xu, Liekang Zeng, Kaiwei Liu, Siyang Jiang, Wenrui Lu, Hongkai Chen, Xiaofan Jiang, Guoliang Xing, and Zhenyu Yan. Contextagent: Context-aware proactive llm agents with open-world sensory perceptions. *arXiv preprint arXiv:2505.14668*, 2025.
- [80] Senqiao Yang, Yukang Chen, Zhuotao Tian, Chengyao Wang, Jingyao Li, Bei Yu, and Jiaya Jia. Visionzip: Longer is better but not necessary in vision language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19792–19802, 2025.
- [81] Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu, Ce Liu, Michael Zeng, and Lijuan Wang. Mm-react: Prompting chatgpt for multimodal reasoning and action. *arXiv preprint arXiv:2303.11381*, 2023.
- [82] Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- [83] Yue Yu, Wei Ping, Zihan Liu, Boxin Wang, Jiaxuan You, Chao Zhang, Mohammad Shoeybi, and Bryan Catanzaro. Rankrag: Unifying context ranking with retrieval-augmented generation in llms. *Advances in Neural Information Processing Systems*, 37:121156–121184, 2024.

- 660 [84] Li-Ming Zhan, Bo Liu, Lu Fan, Jiaxin Chen, and Xiao-Ming Wu. Medical visual question
661 answering via conditional reasoning. In *Proceedings of the 28th ACM international conference*
662 *on multimedia*, pages 2345–2354, 2020.
- 663 [85] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. Bertscore:
664 Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*, 2019.
- 665 [86] Xiaoman Zhang, Chaoyi Wu, Ziheng Zhao, Weixiong Lin, Ya Zhang, Yanfeng Wang, and Weidi
666 Xie. Pmc-vqa: Visual instruction tuning for medical visual question answering. *arXiv preprint*
667 *arXiv:2305.10415*, 2023.
- 668 [87] Xuan Zhang, Yang Deng, Zifeng Ren, See Kiong Ng, and Tat-Seng Chua. Ask-before-plan:
669 Proactive language agents for real-world planning. In *Findings of the Association for Computa-*
670 *tional Linguistics: EMNLP 2024*, pages 10836–10863, 2024.
- 671 [88] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
672 Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and
673 chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.
- 674 [89] Yuxin Zuo, Shang Qu, Yifei Li, Zhangren Chen, Xuekai Zhu, Ermo Hua, Kaiyan Zhang, Ning
675 Ding, and Bowen Zhou. Medxpertqa: Benchmarking expert-level medical reasoning and
676 understanding. *arXiv preprint arXiv:2501.18362*, 2025.

677 A Supplementary Reproducibility and Checklist Details

678 This appendix provides supplementary information needed to support the reproducibility, release,
679 safety, and checklist statements in the main paper. The main text is authoritative for task definition,
680 method design, and reported results.

681 A.1 Benchmark Asset Details

682 MedXR-Bench is constructed from publicly available *NEJM* Image Challenge cases, as described in
683 Section 3.2. The processed corpus contains 1,066 cases, split into 795 training cases and 271 test
684 cases. For each source case, we parse the patient description, diagnostic question, visual evidence,
685 answer-related metadata, and disease-category label. We then construct an outpatient-style trace
686 containing doctor–patient dialogue, synthesized clinical audio, egocentric-style visual observations,
687 structured patient context, and targets for clinical summarization and open-ended diagnosis or etiology
688 prediction.

689 To avoid multiple-choice shortcuts and source-answer leakage, the original answer options, correct
690 answer, and source explanation are excluded from all model-facing inputs and retrieval queries. These
691 fields are retained only as offline metadata for target construction and evaluation. The benchmark does
692 not introduce private patient data: all clinical content is derived from the public source cases, audio is
693 generated by TTS from the constructed dialogue, and no new personally identifying information is
694 added.

695 A.2 Supplementary Benchmark Comparison

696 Table 4 summarizes how MedXR-Bench differs from representative medical VQA datasets, medical-
697 agent benchmarks, and context-aware agent benchmarks. The comparison highlights the benchmark’s
698 focus on continuous egocentric clinical reasoning over streaming audio, first-person visual observa-
699 tions, structured patient context, retrieved evidence, and final clinical summarization plus open-ended
700 diagnosis.

701 A.3 Implementation, Evaluation Settings, and Compute

702 The main pipeline and post-training method are described in Sections 4 and 4.4. Table 5 summarizes
703 the model/API versions used in the reported experiments, and Table 8 reports representative compute.

704 **AR glasses prototype.** The MedXRAgent client is implemented as a Unity-based application and
705 compiled for the device’s native Android operating system. The AR glasses provide a lightweight
706 wearable interface for hands-free, first-person clinical data capture and clinician-facing feedback.
707 Despite the device’s compact form factor of only 76g, it includes dual front-facing cameras for
708 continuous egocentric video recording, as well as built-in microphones and speakers for voice-based
709 interaction during the encounter. The binocular full-color display, powered by a Micro LED optical
710 engine, allows the system to present compact on-demand guidance while preserving the clinician’s
711 natural line of sight and doctor-patient communication. These hardware properties make the device
712 suitable for collecting continuous encounter traces across consultation, bedside examination, and
713 patient repositioning without requiring manual camera operation or disrupting routine workflow.
714 Detailed specifications are summarized in Table 7.

715 For the LLM-as-Judge metrics in Section 5.1, GPT-5.4-mini uses five independent judge calls per
716 sample, and scores are normalized to $[0, 1]$ before median aggregation. The Open-End QA judge
717 evaluates clinical equivalence to the reference answer, support from encounter evidence, and critical
718 contradictions or unsafe failures.

719 A.4 Statistical Reporting

720 The main paper reports aggregate scores in Section 5.2 and ablations in Section 5.3. To distinguish
721 headline improvements from judge stochasticity, we use repeated LLM-as-Judge calls and median
722 aggregation at the case level. Differences smaller than roughly 0.01 are treated as judge noise based
723 on single-run versus median-of-five stability checks.

Table 4: Comparison of medical reasoning benchmarks. MedXR-Bench is designed for continuous egocentric, evidence-grounded clinical reasoning rather than static medical QA alone.

Benchmark	Med. domain	Ego-centric	Streaming audio	Multi-turn dialogue	Visual evidence	Patient context	Web evidence	Clinical summary	Open-ended diagnosis	Continuous agent
VQA-RAD	yes	no	no	no	yes	no	no	no	partial	no
PathVQA	yes	no	no	no	yes	no	no	no	partial	no
SLAKE	yes	no	no	no	yes	no	no	no	partial	no
PMC-VQA	yes	no	no	no	yes	no	no	no	partial	no
MedXpertQA	yes	no	no	no	partial	no	no	no	yes	no
AgentClinic	yes	no	no	yes	no	partial	no	no	yes	partial
ContextAgentBench	no	yes	partial	no	yes	yes	no	partial	no	yes
MedXR-Bench (ours)	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes

Table 5: Model and API versions used in the reported pipeline.

Component	Version / model
Central proprietary reasoner	Gemini-3.1-Pro
Open-source central reasoners	Qwen3.5-2B / Qwen3.5-9B
General baseline	GPT-5.4 with GPT-4o-audio
LLM-as-Judge evaluation	GPT-5.4-mini
GRPO reward judge	Gemini Pro
Medical visual fallback	MedGemma-1.5-4B
Dense retrieval index	FAISS over rag_kb_v2 disease entries

The main comparisons have effect sizes larger than this judge-noise threshold: the best MedXRAgent open-ended diagnosis score exceeds the strongest non-Ours baseline by 0.065; Qwen3.5-9B GRPO improves over its zero-shot counterpart by 0.203 on Open-End QA and 0.257 on the summary judge score; and removing web evidence from the MedXRAgent Gemini-centered pipeline reduces Open-End QA by 0.103. We therefore report judge-stability and effect-size evidence for the main comparisons while avoiding formal p -value claims.

A.5 Supplementary Result Visualizations

Figure 6 provides bar-chart views of the two LLM-as-Judge metrics reported in Section 5.2. Figure 7 visualizes the traditional summarization metrics, and Figure 8 summarizes the evidence-channel drops reported in Section 5.3.

A.6 Qualitative Case Examples

Figures 9 and 10 provide representative case examples that illustrate how retrieved evidence, visual evidence, and patient history can support the final diagnosis, as well as failure modes caused by subtle missing evidence, retrieval noise, and overconfident diagnosis.

Release safeguards include excluding answer options and source explanations from model-facing inputs, documenting intended and out-of-scope uses, avoiding the addition of private patient-identifying information, preserving source access conditions for clinical images, and restricting released materials to research evaluation rather than clinical use.

A.7 Limitations, Broader Impacts, and Human-Subjects Statement

Limitations. MedXR-Bench is a synthesized benchmark and does not reproduce the full noise, interruptions, privacy constraints, or workflow of a live outpatient encounter. Cases are drawn from *NEJM* Image Challenge and may over-represent rare, visual, or diagnostically distinctive cases. ASR errors can distort age, dose, and negation. Frame-selection heuristics can miss subtle but diagnostically critical visual findings. Web search may return outdated, non-guideline, or noisy evidence. LLM hallucination remains possible despite evidence grounding and judge-based evaluation. The evaluation uses an LLM-as-Judge protocol rather than a prospective clinician study, and no prospective clinical validation has been performed.

Broader impacts. Potential positive impacts include improved organization of encounter context, support for evidence-grounded clinical reasoning, and reduced clinician cognitive burden. Potential negative impacts include over-reliance on AI-generated diagnoses, hallucinated or stale recommen-

Table 6: Software / hardware stack used for reported local training and benchmark runs.

Component	Version
OS	Ubuntu 24.04.3 LTS, kernel 6.17.0-20-generic
GPU	4× NVIDIA H200 GPUs (141 GB HBM3e each)
GPU driver	580.126.09
CUDA toolkit	12.8
Python	3.12.13
PyTorch	2.10.0 + cu128
Transformers	5.2.0
PEFT	0.18.1
Accelerate	1.13.0
ms-swift	4.1.2
vLLM	0.19.1



Figure 5: RayNeo X3 Pro AR glasses used as the MedXRAgent clinician-facing prototype interface.

754 dations, privacy risks from clinical context, and premature clinical deployment. MedXRAgent is
755 therefore framed as clinician-facing decision support, not an autonomous diagnostic authority.

756 **Human subjects and IRB.** No crowdsourcing was used and no participants were recruited for the
757 reported benchmark evaluation. Clinician input, if any, was limited to informal expert consultation
758 for system design rather than a participant evaluation study. We have obtained IRB approvals for
759 subsequent studies.

Table 7: AR glasses prototype specifications used for the interactive clinical interface.

Component	Specification
Prototype device	RayNeo X3 Pro AR glasses
Client runtime	Unity application compiled for device-native Android OS
Form factor	Lightweight wearable glasses, 76g
Video capture	Dual front-facing cameras for egocentric recording
Audio interface	Built-in microphones and speakers for voice interaction
Display	Binocular full-color Micro LED optical engine
Clinical use	Hands-free capture and compact clinician-facing guidance

Table 8: Representative compute required for the reported experiments. API-backed inference and judging times can vary with provider latency.

Experiment	Compute setting	Wall-clock time
Qwen3.5-2B GRPO post-training	4× H200	about 8 hours
Qwen3.5-9B GRPO post-training	4× H200	about 26 hours
271-case full pipeline inference	reported pipeline configuration	about 6 hours
LLM-as-Judge evaluation	5 judge calls per case	about 1 hour
Ablation run	one evidence-channel ablation	about 20 minutes

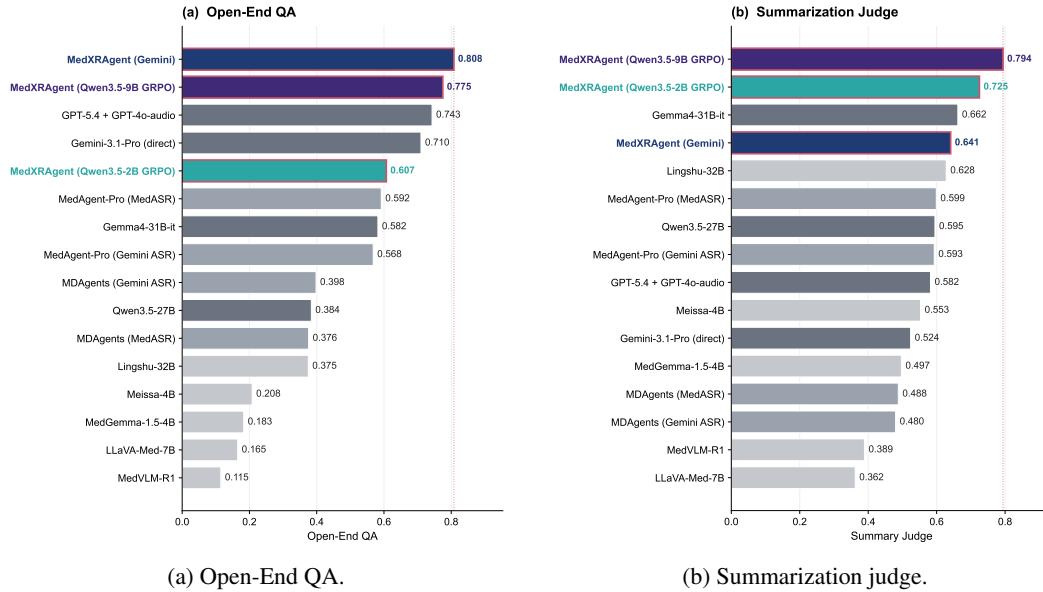


Figure 6: LLM-as-Judge scores for open-ended diagnosis and clinical summarization.

Table 9: Existing assets used by MedXRAgent / MedXR-Bench. Provider API models are governed by provider terms.

Asset	Use	License / terms note
NEJM Image Challenge	Source cases	Permission obtained; source terms apply
Qwen3.5	Open-source reasoner	Qwen release terms
Gemini-3.1-Pro	Proprietary reasoner	Provider API terms
OpenAI GPT models	Audio / judging / baselines	Provider API terms
MedGemma	Medical visual fallback	Model release terms
FAISS	Dense retrieval	Open-source library

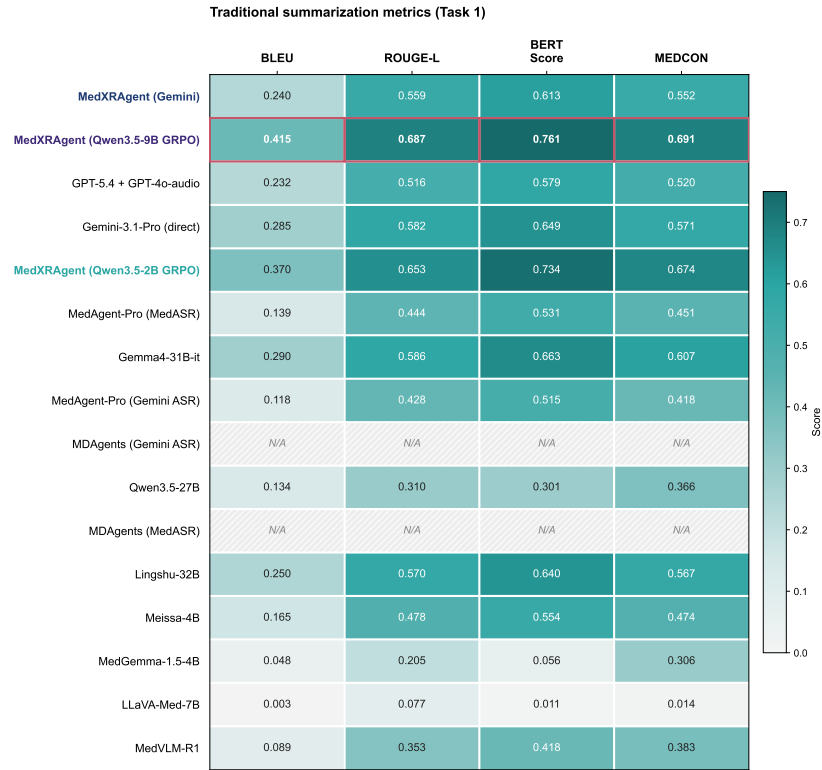


Figure 7: Traditional clinical-summary metrics across evaluated methods.

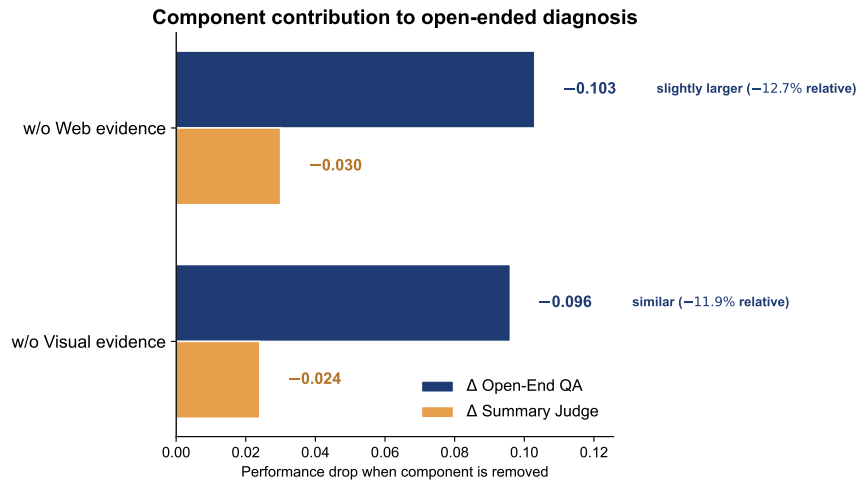
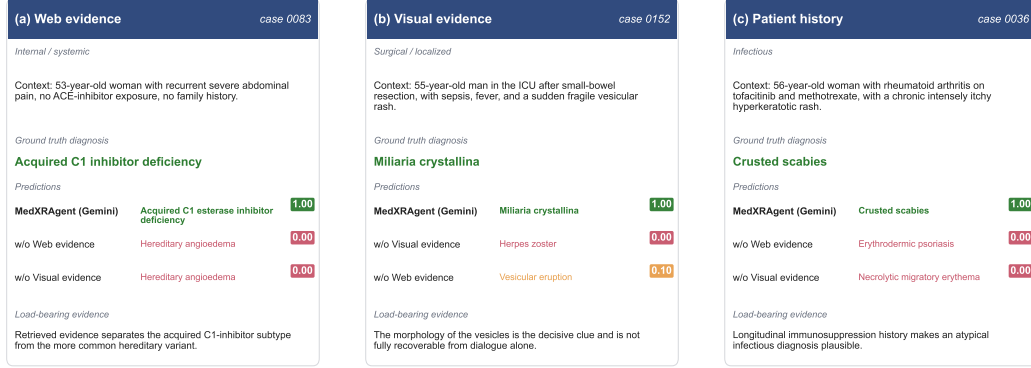


Figure 8: Performance drop when web evidence or visual evidence is removed from the MedXRAgent Gemini-centered pipeline.

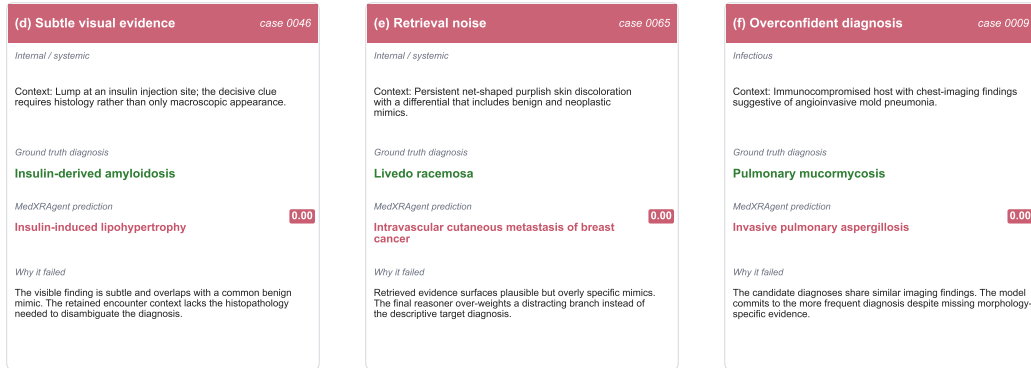


(a) Web evidence.

(b) Visual evidence.

(c) Clinical history.

Figure 9: Representative successful cases where different evidence channels provide load-bearing diagnostic clues.



(a) Subtle visual evidence.

(b) Retrieval noise.

(c) Overconfident diagnosis.

Figure 10: Representative failure cases corresponding to the error modes discussed in Section 5.4.

NeurIPS Paper Checklist

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 5.4 discusses category-specific weaknesses and failure modes, and Appendix A.7 lists limitations including synthetic encounter construction, source-case bias, ASR errors, frame-selection failures, noisy or stale retrieval, LLM hallucination risk, LLM-as-Judge evaluation, and the absence of prospective clinical validation.

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- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
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875 tions to faithfully reproduce the main experimental results, as described in supplemental
876 material?

877 Answer: [Yes]

878 Justification: Section 4.4 states that we release the GRPO recipe, reward plugin, LoRA
879 adapter, and merged Qwen3.5 checkpoint. The release package includes MedXR-Bench
880 manifests, evaluation scripts, judge rubrics, benchmark and ablation code, and release-use
881 constraints.

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905 Answer: [Yes]

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907 system and retrieval configuration, Section 5.1 describes metrics, baselines, and judge setup,
908 and Appendix A.3 gives model versions, hardware/software stack, and runtime settings.

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912 that is necessary to appreciate the results and make sense of them.
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914 material.

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916 Question: Does the paper report error bars suitably and correctly defined or other appropriate
917 information about the statistical significance of the experiments?

918 Answer: [No]

Justification: We do not report error bars or formal statistical significance tests. Error bars are optional for CV-style benchmark papers, and producing them here would be prohibitively costly because it would require rerunning the full multimodal pipeline, post-training, inference, and LLM-as-Judge evaluation across multiple seeds or splits. We therefore report deterministic benchmark scores and ablation effect sizes, and avoid formal p -value claims.

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Answer: [Yes]

Justification: Appendix A.3, Table 6, and Table 8 report the $4 \times$ NVIDIA H200 hardware, software stack, and representative wall-clock runtimes for Qwen3.5-2B GRPO, Qwen3.5-9B GRPO, full-pipeline inference, LLM-as-Judge evaluation, and ablation runs.

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Justification: Sections 4.1–4.4 describe LLM and multimodal LLM roles in dialogue reconstruction, visual evidence use, retrieval-guided reasoning, and GRPO post-training; Section 5.1 and Appendix A.3 report model choices and judge/evaluation settings.

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